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Land Use Land Cover Classification using Machine Learning

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Abstract—***Due to rapid urbanization processes and population exploitation, urban sprawl becomes a challenging task for urban planners. For planning authorities and their decision-making process, particularly in developing nations throughout the late 20th and early 21st centuries, Land Cover Land Use (LULC) is one of the most crucial pieces of information. Instead of doing it manually, the paper's goal is to estimate the LULC Classification using artificial intelligence approaches. The Google Earth Engine (GEE) cloud computing is utilized to easily retrieve satellite photos for this purpose*** [1]***. The suggested approach will make the process of image classification easier so that different land use types may be identified and watched for urbanization. Classification and Regression Trees (CART), a supervised machine learning (ML) technique, is used to perform the classification*** [2]***. Additionally, metrics like classification accuracy, precision and Kappa coefficient are assessed to support the conclusions. The results of the classification revealed a high accuracy of 92.9%. The LULC classification results can be used as a starting point for additional research on a variety of topics, such as river morphology change analysis, ecosystem services analysis, land use policy formulation, management of water resources, management of other natural resources, urbanization, etc.*** [3]***.***

Keywords—Land use land cover (LULC); Classification; Machine Learning; Artificial Intelligence; Regression; Urban Development; Remote Sensing; Classification and Regression Tree (CART); GEE;

I. INTRODUCTION

The amount of woods, wetlands, farmland, other land and water types that cover an area is referred to as its land cover. Land use refers to the way people use the land shows, whether it is for development, conservation, or a combination of the two. When the terms "Land Use" and "Land Cover" are combined, they often refer to the grouping or classification of human activities and natural elements on the landscape throughout a certain time period based on accepted scientific and statistical methods of analysis of relevant source materials [4]. For the study and prediction of many applications, such as weather forecast, agricultural yield, disaster prediction, climate change, urbanization, and so on, the satellite geographical data are of great value. LULC Classification is one such essential use. One of the key tasks in this area is automatically classifying the land based on its physical characteristics and relevant uses. Through LULC classification, different types of land cover can be distinguished with ease to take appropriate action. Most importantly, it is wanted for LULC type future projection for sustainable habitation and development.

The Google Earth Engine (GEE), with its huge reusable library, abundance of geo spatial datasets, and user-friendly interface, is the ideal cloud-based platform for in-depth monitoring and modelling of the features of the earth [5]. GEE has evolved into a strong and efficient platform for picture categorization utilizing satellite photos that can be used for a number of purposes thanks to the availability of free images, tools, and methods for spatial data analysis. One of the various language programming libraries that GEE offers is the JavaScript/Python client library. These libraries can be used to create GEE applications and tool-kits, which can make using GEE easier. GEE also support the use of different machine learning methods.

II. RELATED STUDY

It is easy to combine machine learning algorithms with other open-source network apps and they are a powerful collection of data-driven techniques that employ an automated way for pattern detection in large-scale data [6]. A few of the effective machine learning techniques that have been developed for the detection and modelling of many characteristics are the support vector machine (SVM), random forest (RF), artificial neural network, classification and regression tree (CART), and decision tree [7]. Important algorithms like bagged decision trees, random forests, and boosted decision trees all have their roots in the CART algorithm. A randomly chosen remote sensing sample is used by the CART algorithm to build a binary tree, and the tested sample is then used to prune the tree. The classification accuracy and fast performance of CART make it one of the widely used LULC classifiers. In the research paper released in 2016, A. Sungheetha et al. [8], to maximize efficiency and classification accuracy, the fewer data required the better. Their proposed CNN model is a more filter contained model is that it outperformed the deep model in terms of fine-tuning.

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III. OBJECTIVES

The main objective of this study is to create highly accurate LULC categorization maps of Mumbai City [3].

- To look at the extension and layout of the urban area in the research area [9].
- To assess how land use and cover have changed over time in the study area.
- For LULC classification, cloud computing platforms like GEE can work quickly and effectively.

IV. METHODOLOGY

In this study, the LULC classification is implemented by employing GEE in conjunction with the machine learning algorithm CART, and the sample points of the study region are divided into two parts: the training samples (Fig. 4) and the validation/testing samples (Fig. 5) [6]. The predictive model was created using training data, and the testing samples were used to assess the model's effectiveness [6]. Fig. 1 represents the full methodology [10].

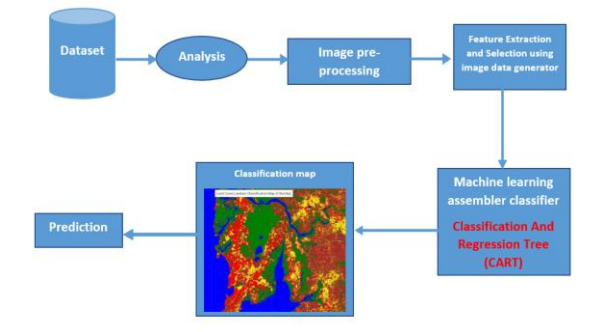

Fig. 1. Detailed Flow of Methodology

A. Study Area

The study area focuses on Mumbai city of India with a total area of about 603.4 sq. Km. Firstly, Mumbai is the financial capital of India and an urban landscape of the western part of the country on coordinate of latitude and longitude between 19.0760° N, 72.8777° E [9]**.** Its central location in terms of connection by land, sea, and air making it a significant industrial and commercial hub. It is the hub of manufacturing, financial services, and capital markets, and it is essential to the expansion and advancement of the Indian economy. Secondly, Mumbai is both one of the world's most populous urban areas and the fourth-most populous city in the world.

B. Machine learning algorithm

One of the supervised machine learning techniques used for classification and regression predictive modelling issues is the classification and regression tree (CART) algorithm [11]. A foundation for crucial algorithms like bagged decision trees, random forests, and boosted decision trees is provided by CART, which runs on the binary decision tree as its basis as shown in Fig. 2. The training set for the decision tree is split in half using the best attribute and threshold value, using the root node as the training set. Identification and tree construction are carried out using training samples. According to a threshold value for an attribute, the nodes are divided into sub-nodes. Additionally, the subsets are divided according to the same logic. This continues until the tree has its final pure sub-set or has all of its potential leaves.

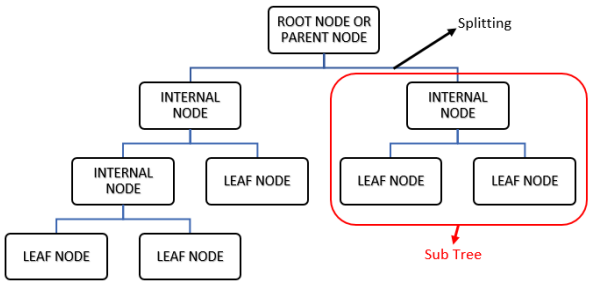

Fig. 2. Binary Decision Tree with CART Algorithm

V. EXPERIMENTAL RESULT

In this section, the Classification And Regression Tree (CART) technique is used to conduct experiments for the classification of LULC on the GEE cloud computing platform [6]. The CART machine learning algorithm was used to process these training and test datasets in Fig. 6, and the results showed a maximum accuracy of 92.9% and precision of 0.92999. The produced LULC map Fig. 3., shows classification of water body, built-up areas, non-built areas, trees, bare land, mangroves using LULC mapping for satellite observations of Mumbai.

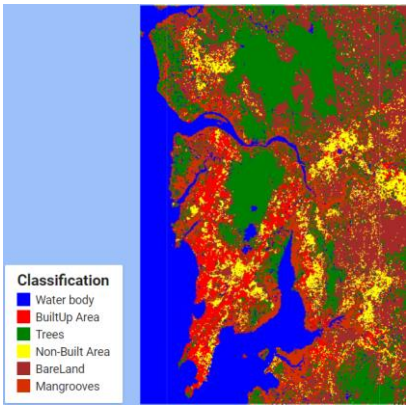

Fig. 3. LULC Classification

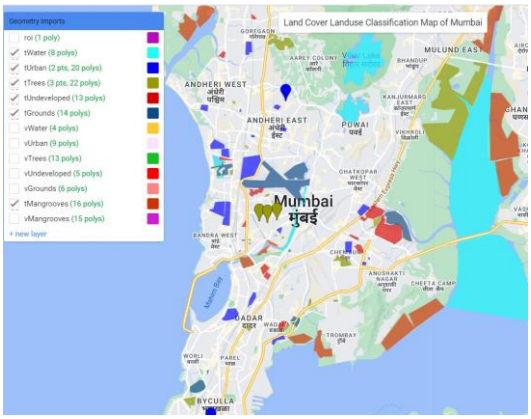

Fig. 4. LULC map for training

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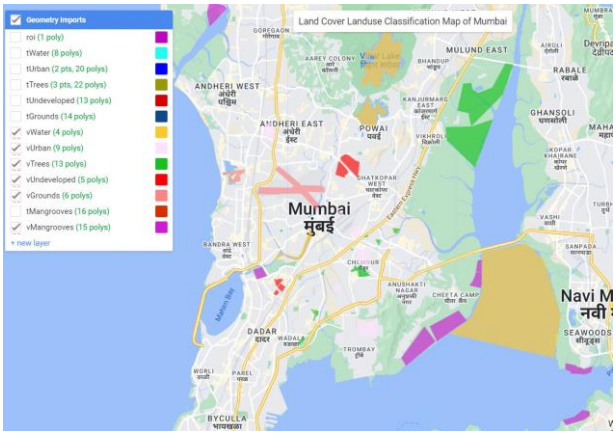

Fig. 5. Validation of LULC

Fig. 6. Dataset geometry imports

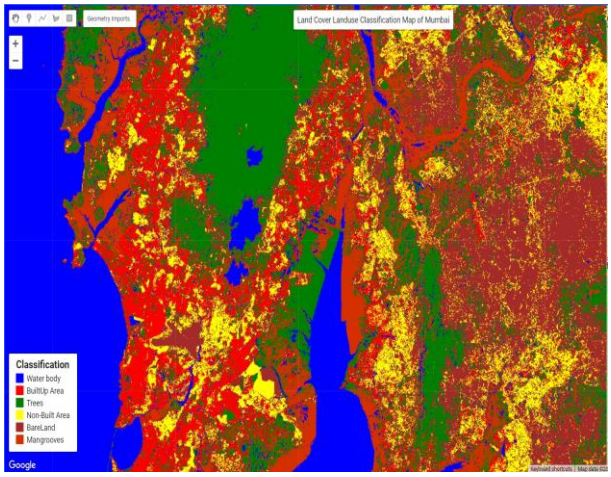

Fig. 7. LULC classification map

The validation error matrix Table I. is a matrix used to determine the performance and the errors in the classification model in the form of a matrix for a given set of data.

TABLE I. VALIDATION ERROR MATRIX

	Water Body [0]	Built-up Area [1]	Trees [2]	Non-Built (Slums) [3]	Grounds/Bare Lands [4]	Mangroves [5]
Water Body [0]	**34049**	29	65	0	0	241
Built-up Area [1]	59	**2158**	211	513	518	67
Trees [2]	422	45	**13390**	5	6	288
Non-Built Area (Slums) [3]	2	7	0	**1312**	200	0
Grounds/ Bare Lands [4]	9	55	51	29	**3004**	3
Mangroves [5]	1056	4	1259	1	4	**13726**

Comparing categorization levels with actual geographic data is known as accuracy assessment. The overall accuracy (in percentage) using Kappa coefficient (K) for classification is shown in Table II. The overall calculated accuracy of LULC classification using CART algorithm has been detected as 93.21% with Kappa coefficient of 0.88 and can be recommended as the best suited classifier of LULC. In the Table II., given below we calculated the Omission Error (OE), Producers Accuracy (PA), Commission Error (CE) and the Users Accuracy(UA) using the error matrix.

TABLE II. ACCURACY ASSESSMENT USING ERROR MATRIX

	[0]	**[1]**	**[2]**	**[3]**	**[4]**	**[5]**	**Total**	CE	UA
[0]	**34049**	29	65	0	0	241	**34384**	0.97%	99.03%
[1]	59	**2158**	211	513	518	67	**4076**	47.06%	52.94%
[2]	422	45	**13390**	5	6	288	**14096**	5.01%	94.99%
[3]	2	7	0	**1312**	200	0	**1521**	13.74%	86.26%
[4]	9	55	51	29	**3004**	3	**3191**	4.61%	95.39%
[5]	1056	4	1259	1	4	**13726**	**16050**	14.48%	85.52%
Total	**35597**	**2298**	**14976**	**1860**	**3772**	**14815**	**73318**	0.97%	99.03%
OE	4.35%	6.09%	10.59%	29.46%	19.30%	7.35%	4.35%	6.09%	10.59%
PA	95.65%	93.91%	89.41%	70.54%	80.70%	92.65%	95.65%	93.91%	89.41%

Cohen's kappa measures the agreement between two raters who each classify N items into C mutually exclusive categories [12]. The equation for KC is:

$$KC = \frac{p(a)-p(r)}{1-p(r)} \quad (1)$$

where $p(a)$ is the Overall Classification Accuracy, and $p(r)$

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is the Random Accuracy. The correctly categorised areas are represented by the diagonal elements in the error matrix Table I. The overall accuracy is determined by adding the number of sites that were correctly categorised and dividing it by the total number of reference sites. Random Accuracy is calculated by summation of ratio of each reference site to total reference cites multiplied by the total of ratio of classification sites to the total reference sites. The Kappa Coefficient may range between -1 and 1 [13]. A value closer to 0 means that the classification is no more accurate than a random one. A negative value means the classification is considerably less accurate than random. The categorization is clearly superior to random when the value is near to 1.

Overall Classification Accuracy $p(a)$: 0.92

Random Accuracy $p(r)$: 0.315

Kappa Coefficient(KC): = 0.88

VI. CONCLUSION

It is commonly accepted that LULC maps and the changes they represent can be taken into account for a variety of applications, including land use planning and allocation, analyses of environmental effect, and assessments of sustainable development [14]. From an environmental perspective, the conclusions are essential for decision-makers and authorities to analyze LULC changes and develop the proper environmental protection strategies [6]. Mumbai's LULC classification findings revealed an overall accuracy of 92.9% (Fig. 7). Because there were many variables taken into account in this study we were faced with a high-dimensional feature space, and hence feature selection approaches are advised for the future studies. Future work in LULC research should address how best to leverage satellite-based land cover databases through LULC change modeling techniques. This will make important contributions to the emerging fields of research in environmental monitoring, global warming, and climate change [15].

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